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DBMS DA-1

Project Title :

Railway Management System

Problem Statement :

The Railway Reservation System is developed to manage train schedules, ticket booking, passenger details, and station information in an efficient and organized manner.
Manual reservation systems are slow, error-prone, and difficult to manage when handling large numbers of passengers, trains, and routes. This system provides a computerized solution that stores all railway-related data such as user details, train information, ticket status, seat availability, and station schedules in a centralized database.
The system allows users to book and cancel tickets, check train availability, and view journey details. Railway authorities can manage train schedules, station information, and ticket status efficiently.

Objectives :

- To manage user and passenger details
- To store train schedules and seat availability
- To handle ticket booking and cancellation
- To maintain accurate reservation records
- To reduce manual errors

System Modules :

1. User Management Module
This module handles user registration, login, and profile management.
It stores user details such as name, address, contact number, email, and security credentials.
Users can book and cancel tickets through this module.
Entity used: USER

2. Train Management Module
This module maintains train information including train number, train name, source, destination, arrival time, departure time, and seat availability.
Railway authorities can update train schedules using this module.
Entity used: TRAIN

3. Train Status Module
This module manages seat availability and fare details for different classes such as AC, Sleeper, and General.
It helps users check the number of available seats and ticket prices.
Entity used: TRAIN_STATUS

4. Station Management Module
This module stores information about railway stations such as station number, name, arrival time, and halt time.
It tracks where trains start, stop, and reach.
Entity used: STATION

5. Ticket Management Module
This module handles ticket booking, ticket status, number of passengers, and journey details such as source and destination stations.
It connects users, passengers, and trains.
Entity used: TICKET

6. Passenger Management Module
This module stores passenger details such as name, age, gender, seat number, and reservation status.
Each passenger is linked to a ticket.
Entity used: PASSENGER

7. Payment Management Module
This module manages ticket payment information including payment amount, mode (UPI, card, cash), payment status, and date.
Each ticket has one payment record.
Entity used: PAYMENT

8. Seat Management Module
This module handles seat allocation for passengers.
It stores coach number, seat number, seat type, and availability for each train.
Entity used: SEAT

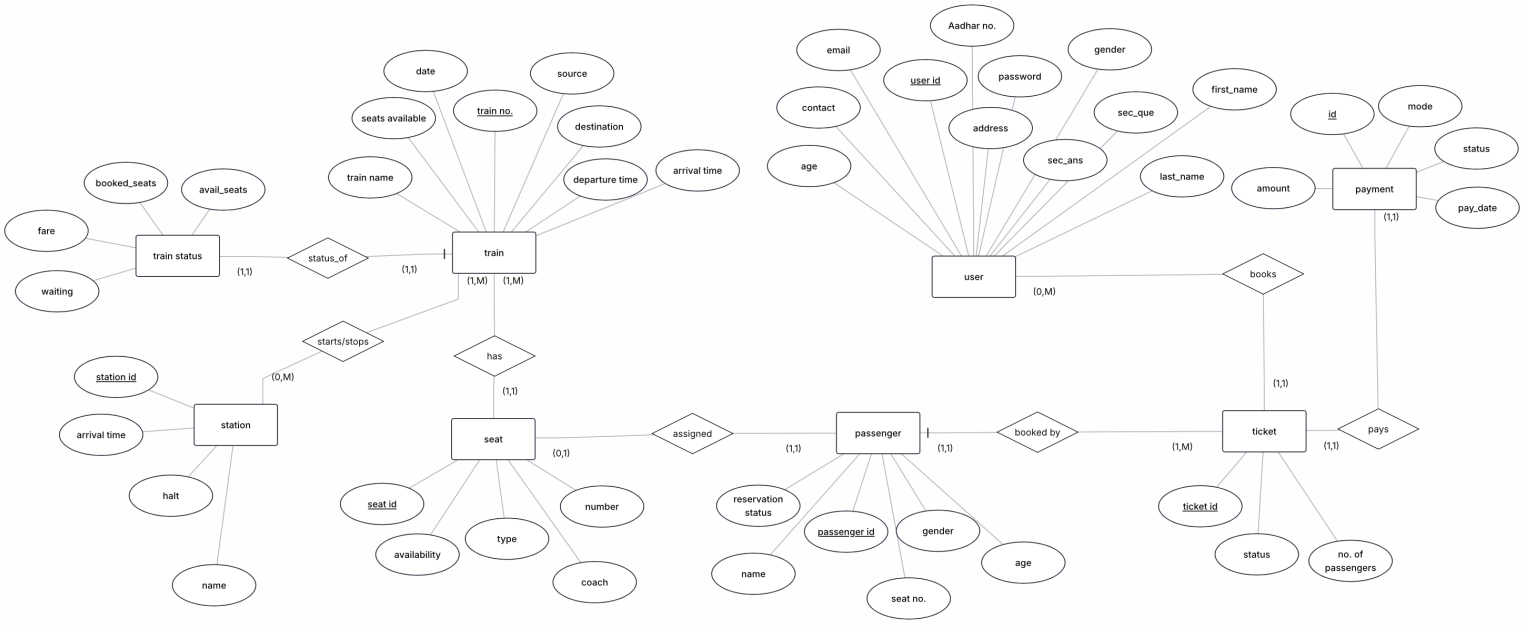
Entities and Attributes

Entities	Attributes
User	user_id (PK) password first_name last_name address mobile_no email aadhar_no age gender security_ques security_ans
Train	train_no (PK) train_name source destination departure_time arrival_time availability_of_seats date
Train status	a_seats1 a_seats2 b_seats1 b_seats2 w_seats1 w_seats2 fare1 fare2
Station	no (PK) name arrival_time halt
Ticket	id (PK) no_of_passengers status booked_user

Entities	Attributes
Passenger	passenger_id (PK) name pnr_no age gender seat_number reservation_status
Payment	payment_id (PK) ticket_id (FK) amount payment_mode payment_status payment_date
Seat	seat_id (PK) train_no (FK) coach seat_number seat_type availability

Er diagram :

Made using online website erdplus.com



Relational Schema for Railway Management System

The ER diagram is mapped into the following relational schema.
Each entity is converted into a table.
Primary keys uniquely identify records, and foreign keys represent relationships.

USER

User id	password	First name	last_name	address	Contact
Email	Aadhar no.	Age	Gender	sec_que	sec_ans

TRAIN

Train no.	Name	souce	Destination
Arrival time	Departure time	Date	seats_available

TRAIN_STATUS

Train no.	avail_seats	booked_seats	Waiting	Fare

STATION

Station id	Arrival time	Halt	Name

TICKET

Ticket id	Status	No. Of passengers	User id

PASSENGER

Passenger id	Name	Gender	Seat no.	Reservation status	Ticket id

PAYMENT

id	Mode	Status	pay_date	Ticket id	Amount

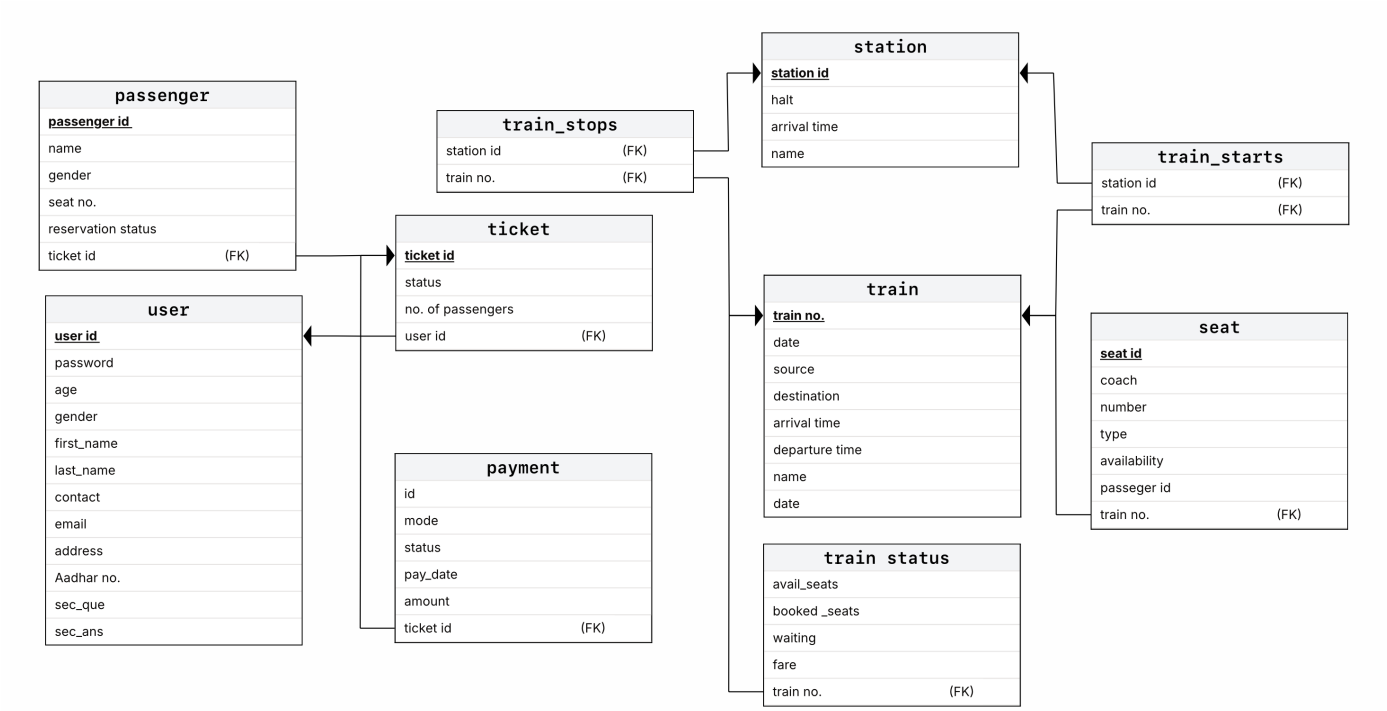
SEAT

Seat id	Coach	Number	Type	Availability	Train no.	Passenger id

TRAIN_STARTS

Train no.	Station id

TRAIN_STOPS

Train no.	Station id

Schema Diagram :